

1. Calculate

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|---|---|
| (a) $\int x \arcsin x dx.$ | (n) $\int (\sin 3x - \sin x)^2 dx.$ |
| (b) $\int \frac{x}{(x+1)(x+2)(x+3)} dx.$ | (o) $\int \frac{1}{1+\sin x+\cos x} dx.$ |
| (c) $\int \frac{x+2}{\sqrt{x^2+4x+13}} dx.$ | (p) $\int \frac{\sec^2 \theta}{\tan^3 \theta - \tan^2 \theta} d\theta.$ |
| (d) $\int \frac{x^2+1}{x(x^2-1)} dx.$ | (q) $\int x^3 \arctan x dx.$ |
| (e) $\int \frac{\sqrt{x^2-1}}{x} dx.$ | (r) $\int \csc^2 2x dx.$ |
| (f) $\int \sin^3 x \cos^2 x dx.$ | (s) $\int \cos 2x \sin 3x dx.$ |
| (g) $\int \frac{dx}{(x^2+16)^2}.$ | (t) $\int \tan^3 x \sec^3 x dx.$ |
| (h) $\int \frac{1}{(x-1)(x^2+1)^2} dx.$ | (u) $\int \frac{2x^2}{(4x+1)^{5/2}} dx.$ |
| (i) $\int \sin^2 x \sin 2x dx.$ | (v) $\int \frac{1-e^x}{1+e^x} dx.$ |
| (j) $\int \frac{x^2-4x+3}{x^3+2x^2+x} dx.$ | (w) $\int \frac{\sin x}{1+\sin^2 x} dx.$ |
| (k) $\int \frac{dx}{\sqrt{x^2-2x-3}}.$ | (x) $\int \cos(\ln x) dx.$ |
| (l) $\int \frac{\cos \theta}{\sin^2 \theta - 2 \sin \theta - 8} d\theta.$ | (y) $\int (e^x + 2x)^2 dx.$ |
| (m) $\int \frac{1}{t(\ln t)^2 - 4} dt.$ | (z) $\int \frac{\arcsin 2x}{\sqrt{1-4x^2}} dx.$ |

2. The region under the curve $y = 1/(1+x^2)$ from $x = 0$ to $x = 1$ is revolved about the x -axis. Find the volume of the resulting solid.

3. Show that for $y = \frac{1}{x^2-1}$,

$$\frac{d^n y}{dx^n} = \frac{(-1)^n n!}{2} \left[\frac{1}{(x-1)^{n+1}} - \frac{1}{(x+1)^{n+1}} \right].$$

4. Calculate $\int \frac{1}{x} \sqrt{a^2 - x^2} dx.$

- (a) By setting $u = \sqrt{a^2 - x^2}.$
 (b) By a trigonometric substitution.
 (c) Then reconcile the results.